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Digital Ticketing System for Ceres Liner in Negros Occidental

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Chapter 1

Background of the Study:

In recent times, the growing demand for faster, more convenient services has pushed many industries including transportation to rethink how they operate. At Ceres Liner in Negros Occidental, the traditional ticketing system still relies on manual processes, which often lead to long lines, booking mistakes, and disorganized passenger data (Adducul & Adducul, 2020). These issues not only slow down operations but also affect the overall experience of passengers who expect quicker and smoother service.

Based on national and global studies, many organizations have seen improvements after shifting to digital solutions. For example, Cendana and Bustillo (2018) found that smart ticketing systems help reduce errors and make travel more convenient for customers. Similarly, Barabino, Lai, and Olivo (2020) showed that digital systems can improve fare collection and create more transparent, reliable operations. These studies show how technology can make a real difference in public transportation.

With this in mind, the researchers aim to create a digital ticketing system specifically for Ceres Liner. The goal is to use technology to simplify the booking process, avoid common errors, and make things easier for both staff and passengers.

This study will focus on building a prototype of an online ticket booking and processing system. It will look into how well the system works, how much it can reduce manual work, and how it can improve customer satisfaction. In doing so, the project hopes to support the broader move toward digital transformation in the transportation sector.

Statement of the Problem:

In today's fast-paced world, transportation services are expected to provide fast, reliable, and convenient solutions for passengers. However, Ceres Liner in Negros Occidental still uses a manual ticketing system, which often leads to booking errors, long queues, and inefficient data management. These issues affect both operations and customer satisfaction. The main objective of this study is to design and develop a digitalized ticketing system for Ceres Liner in Negros Occidental to address these challenges:

1. **Inefficiency in Managing Ticket Issuance:** Manual ticketing processes lead to longer transaction times, increased human errors, and difficulty in tracking issued tickets, causing delays and revenue management issues.

2. **Lack of Effective Data Management and Reporting**: The absence of a digital system limits the ability to generate real-time reports, track ticket sales, and manage financial data effectively, leading to operational inefficiencies.

3. **Payment Issues** :Negros Ceres mainly uses cash for payments, causing delays, mistakes, and problems in tracking money. Although GCash is an option, not all passengers use it. Without a better payment system, it is harder to serve passengers quickly and keep track of payments accurately. A system that accepts both cash and GCash can make payments easier and faster.

Statement of Objective

This study aims to design and develop a digitalized ticketing system for Ceres Liner in Negros Occidental. It seeks to improve the efficiency of ticket booking, reduce human errors, and streamline data management. The system will also support both cash and GCash payments to provide more flexible and faster transaction options. Ultimately, the project aims to enhance the overall experience for both passengers and staff through a more reliable and modern ticketing process.

1. To design and develop a digitalized ticketing system for Ceres Liner that automates the booking and ticket issuance process, reducing manual errors and long queues

2. To improve data management by enabling real-time tracking of ticket sales, generating reports, and organizing passenger records efficiently.

3. To integrate a dual payment option system that accepts both cash and GCash to speed up transactions and enhance customer convenience.

Scope and Limitation of the Study

This research focuses on the development and implementation of a digitalized ticketing system for Ceres Liner. The study includes:

1. **Payment Management**:
   - Support for both cash and GCash payments.
2. **User Roles and Access:**
   - Separate access levels for administrators, conductors, and passengers.
3. **Ticket Management:**
   - Automated ticket booking and issuance.

However, the study is limited to:

1. **Payment Methods:**
   - While the system will support GCash and cash payments, it will not integrate with other digital payment platforms like credit cards or bank transfers in the initial phase.

2. **Internet Dependency**:
   - Some system functions, like real-time reporting and payment verification, may require an internet connection. Offline operations will be limited.

3. **Scope of Operations:**
   - The system is designed for the ticketing and payment processes of Negros Ceres and may not support third-party or partner services

Significance of the study

As transportation needs grow, traditional ticketing methods are no longer enough. This study aims to develop a digital ticketing system for Ceres Liner in Negros Occidental to reduce booking errors, shorten waiting times, and improve data management.

This study matters for a few key reasons:

Ceres Liner: The new system will help speed up ticketing, cut down on mistakes, and make it easier to track ticket sales and finances, which will make day-to-day operations much more efficient.

Employees: By organizing the system with different roles for staff, it will make their jobs clearer and easier, helping reduce stress and confusion.

Passengers: The system will make buying tickets faster and offer more payment options, like cash and GCash, so passengers can get through the process quickly and easily.

Other Transport Companies: This study can also serve as a helpful example for other transportation companies still using old manual systems, showing them how switching to digital can improve service.

Community: A more modern and efficient system means better public transportation for everyone, which contributes to the overall development of the local area.

This study aims to make travel easier for passengers and help Ceres Liner run better by using a digital system. It is a step toward improving public transportation for everyone.

Definition of Terms

Administrator
Conceptual: A person who manages and controls a system.
Operational: The person who controls the ticketing system, updates information, and checks sales.

Booking
Conceptual: Reserving a seat, ticket, or service ahead of time.
Operational: When a passenger reserves a bus seat using the digital ticketing system.

Cash Payment
Conceptual: Paying money in coins or bills for a service or product.
Operational: A way for passengers to pay for tickets directly using cash.

Conductor

Conceptual: A bus staff member who checks and manages tickets.

Operational: The staff who checks if the passenger has a valid ticket using the system.

Customer Satisfaction

Conceptual: How happy customers are with a service or product.

Operational: How passengers feel about using the new ticketing system.

Data Management

Conceptual: Organizing and maintaining data to keep it accurate and accessible.

Operational: The system saves and arranges passenger details, ticket sales, and reports.

Digital Ticketing System

Conceptual: A system that handles ticket booking and payments electronically.

Operational: The system used by Ceres Liner to book tickets, give tickets, and accept payments online.

E-Receipt

Conceptual: An electronic proof of payment.

Operational: The digital receipt given to passengers after booking and paying.

Fare Collection

Conceptual: Gathering payment for transportation services.

Operational: The system that collects payments through cash or GCash.

GCash

Conceptual: A mobile wallet app in the Philippines for sending, receiving, and spending money.

Operational: A way for passengers to pay using their phones instead of cash.

Internet Connection

Conceptual: A service that allows devices to access the web.

Operational: Needed by the system to update bookings, payments, and reports in real-time.

Manual Ticketing

Conceptual: Issuing tickets and recording sales manually by hand.

Operational: The current way Ceres Liner gives paper tickets and writes down sales by hand.

Mobile Payment

Conceptual: Paying using a smartphone or tablet instead of cash or card.

Operational: Paying for bus tickets using GCash on a phone.

Passenger

Conceptual: A person who travels by bus or other transportation.

Operational: The customer who books a seat and uses the digital ticketing system.

Payment System

Conceptual: A way to transfer money between a buyer and a seller.

Operational: How the system lets passengers pay using cash or GCash.

Queue

Conceptual: A line of people waiting for a service.

Operational: What the system aims to reduce by allowing online ticket booking.

Real-Time Reporting

Conceptual: Getting updated data instantly.

Operational: The system shows ticket sales, payments, and bookings right away.

Revenue Management

Conceptual: Managing money coming into a business.

Operational: Tracking ticket payments and total sales automatically through the system.

Seat Availability

Conceptual: Checking if seats are open for booking.

Operational: The system shows which seats are already taken or still available.

User Roles

Conceptual: Different levels of access and tasks given to users in a system.

Operational: The system gives different access: admins manage settings, conductors check tickets, and passengers book tickets.

Chapter 2

Review of related literature and studies

As technology continues to revolutionize service industries, public transportation has also adapted to digital innovations to improve operational efficiency and customer experience. The traditional manual ticketing system, while still in use in many areas such as Negros Occidental, has shown limitations in terms of speed, accuracy, and data handling. In light of these challenges, numerous studies and literature both locally and internationally have examined the shift towards digital ticketing systems. These works explore system design, user adoption, payment integration, and overall impact on service quality. This chapter presents relevant literature and studies that serve as a foundation for developing a digital ticketing system tailored to the needs of Ceres Liner, highlighting both foreign and local perspectives from 2021 to 2025.

Foreign related literature

A Reusable Public Transport Electronic Ticket System with Fast Validation (Borges & Sebé, 2025)

Borges and Sebé (2025) introduced an innovative electronic ticketing system designed for public transportation, emphasizing both reusability and rapid validation. The system allows travelers to use a single ticket for multiple journeys within a specified time frame, enhancing convenience and reducing the need for multiple ticket purchases. A notable feature of their system is its commitment to user privacy; travelers can validate tickets without disclosing personal information, ensuring anonymity and unlinkability. This is achieved through advanced cryptographic techniques that secure transactions and protect user data.

The relevance of this study to the proposed digital ticketing system for Ceres Liner lies in its focus on efficiency and privacy. By adopting similar cryptographic methods, Ceres Liner can offer passengers a seamless and secure ticketing experience, minimizing boarding times and enhancing overall service quality. Furthermore, the system's ability to function effectively with minimal delays makes it suitable for high-traffic scenarios, ensuring that the transportation process remains smooth and efficient.

Digital Payment Adoption in Public Transportation: Mediating Role of Mode Choice Segments (Madurapperuma et al., 2025)

Madurapperuma et al. (2025) conducted a comprehensive study on the adoption of digital payment systems in public transportation within developing cities. Utilizing a two-step methodology involving Latent Class Cluster Analysis and machine learning algorithms, the study segmented travelers based on socio-demographic factors and travel behaviors. Key findings highlighted that factors such as prior use of travel apps, smartphone ownership, internet accessibility, and age significantly influence the likelihood of adopting digital payment methods.

The insights from this study are particularly pertinent to the implementation of a digital ticketing system for Ceres Liner. Understanding the diverse preferences and technological readiness of different passenger segments can guide the design of a user-friendly system that accommodates both tech-savvy users and those less familiar with digital tools. By tailoring the system to address these varied needs, Ceres Liner can enhance user adoption rates and ensure a more inclusive transportation service.

Local related literature

Who Uses Smart Card? Understanding Public Transport Payment Preference in Developing Contexts: A Case Study of Manila's LRT-1 (Castro & Baluyot, 2023)

Castro and Baluyot (2023) explored the factors influencing the adoption of smart card payment systems among commuters of Manila's LRT-1. Their study revealed that cultural practices, such as the "tingi-tingi" or sachet economy, play a significant role in shaping payment preferences. Many commuters preferred single journey tickets over smart cards due to daily budgeting habits and limited disposable income. Additionally, the study identified that the perceived complexity of smart card systems and lack of awareness hindered widespread adoption.

These findings underscore the importance of considering local cultural and economic contexts when implementing digital payment systems. For Ceres Liner, this implies that while introducing a digital ticketing system, it's crucial to maintain options that cater to passengers who may be hesitant or unable to transition to digital methods immediately. Providing clear instructions, user education, and maintaining traditional payment options during the transition phase can facilitate smoother adoption and ensure that no passenger segment is left behind.

Crowdsourcing and Collaborative Governance in General Santos City (Padua & De Leon, 2023)

Padua and De Leon (2023) examined the role of crowdsourcing and collaborative governance in enhancing public transportation systems in General Santos City. Their research highlighted how engaging citizens through digital platforms allowed for real-time reporting of transportation issues, leading to more responsive and effective governance. The study emphasized that incorporating community feedback into transportation planning resulted in services that better met the needs of the populace.

Applying these insights to the Ceres Liner context suggests that integrating feedback mechanisms within the digital ticketing system can be highly beneficial. By allowing passengers to report issues, suggest improvements, and provide feedback directly through the platform, Ceres Liner can foster a sense of community involvement and continuously refine its services based on actual user experiences. Such participatory approaches can lead to increased passenger satisfaction and more efficient transportation operations.

Synthesis for related literature

The reviewed literature highlights the growing significance of digital ticketing systems in transforming public transportation, both globally and locally. Foreign studies, such as those by Borges and Sebé (2025) and Madurapperuma et al. (2025), emphasize the advantages of implementing reusable and secure digital ticketing systems, particularly those that prioritize fast validation and user privacy. These systems not only streamline fare collection but also improve operational efficiency by reducing manual processes and boarding times. Moreover, they showcase how socio-demographic factors and smartphone accessibility influence the adoption of digital payments, providing a guide for inclusive and user-friendly design.

Similarly, local literature by Castro and Baluyot (2023) and Padua and De Leon (2023) stresses the need to adapt digital solutions to cultural practices and the importance of community involvement in system development. Commuters' habits, financial behavior, and digital literacy significantly affect the acceptance of new technologies like smart cards or mobile payment apps. Furthermore, collaborative governance and crowdsourced feedback mechanisms can improve transportation services by aligning digital system features with user expectations and real-time needs.

A common theme across all the reviewed sources is the need for a hybrid approach—balancing advanced digital functionality with accessibility for users who may not be digitally literate. Providing both cash and mobile payment options, simplifying the user interface, and offering multilingual support are all strategies suggested to make the system more inclusive. The integration of analytics and real-time reporting is also seen as a key factor in enabling transport companies to optimize services based on data-driven insights, further enhancing system efficiency and transparency.

In summary, both foreign and local literature converge on the value of digital transformation in public transportation, highlighting efficiency, user-centered design, and community participation as key elements of successful implementation. These insights strongly support the development of a digital ticketing system for Ceres Liner that is inclusive, adaptive, and efficient—geared toward solving long-standing issues like manual errors, long queues, and limited payment options.

Foreign related studies

Mobile Bus Ticketing System: Development and Adoption (Ramonsito B. Adducul & Ian Monel C. Adducul, 2020)

This study focused on the creation and implementation of a mobile bus ticketing system intended to address the inefficiencies in traditional manual ticketing in public transportation. It evaluated the benefits of adopting mobile solutions, especially in minimizing human errors during booking and payment processes. The researchers found that digitization significantly enhanced operational speed and reduced the complexity of fare collection.

The study also emphasized user satisfaction, noting that commuters appreciated the reduced waiting times and improved accessibility. The research demonstrated how the integration of a digital ticketing platform enabled real-time tracking of sales and passenger data, leading to better revenue management. The authors recommend wider adoption of similar systems, especially for bus lines that serve a high volume of passengers daily.

Advanced Digital Ticketing and Payment Solutions in Smart Transportation Systems (Avasthi et al.,2025)

This case study investigated the implementation of digital ticketing in Kigali Bus Services, aiming to address fare collection challenges and outdated ticketing operations. The findings showed that the system improved fare transparency and minimized financial leakages due to fraud and manual accounting. Moreover, the automated system helped streamline the entire booking process and reduced operational delays.

It is also noted that training employees to use the system played a vital role in successful adoption. The study further highlighted that digital ticketing is not only a technological upgrade but also a strategic shift toward sustainable public transport systems. Overall, the study supports the idea that modern digital platforms can effectively solve traditional inefficiencies in transport services.

Local related studies

A Web-based Ticketing and Booking Application for Enhancing Maritime Travel Operations and Services in Siargao Island (Riah Encarnacion, 2024)

This research developed a web-based ticketing and booking platform tailored for maritime transportation in Siargao. The system aimed to address several recurring problems in port operations such as long queues, manual data entry, and inconsistent scheduling. By shifting to an online system, the application offered passengers real-time booking, fare estimation, and electronic receipts, improving customer experience.

The researchers found that the new system improved both service delivery and internal reporting processes. Employees were able to generate sales reports quickly, while passengers appreciated the convenience of booking remotely. The study concluded that such systems could be scaled for use in land transportation, especially in regions with a high volume of daily commuters like Negros Occidental

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Impact of Digital Platform Economy and Ride-Hailing Services on Urban Mobility in Pampanga (Galang et al., 2025)

This study explored how ride-hailing apps and digital platforms affected commuting in urban areas like Pampanga. The researchers examined factors such as convenience, travel time, and commuter preferences, finding that users preferred app-based booking systems over manual hailing due to speed and reliability. It demonstrated that digital mobility platforms contribute significantly to reducing urban congestion and optimizing route planning.

Moreover, the study emphasized that integrating mobile wallets and online payments (e.g., GCash) improved fare collection transparency. The authors noted that digital platforms created not just convenience for passengers, but also new job opportunities for transport operators. These findings are particularly relevant, as they underscore the value of adopting multi-payment support in modern transport systems like Ceres Liner.

Synthesis for related Studies

The transition from traditional manual ticketing systems to digital ticketing platforms has been a significant trend in the transportation sector globally and locally, addressing key inefficiencies such as long queues, human errors, and poor data management. Foreign studies like those by Adducul & Adducul (2020) and Avasthi et al**.** (2021) demonstrate that mobile and digital ticketing systems significantly improve operational efficiency, reduce errors in fare collection, and provide better customer satisfaction. Adducul & Adducul's study found that mobile systems could streamline booking and payment processes, making transportation more convenient for passengers, while Avasthi et al. research emphasized the improved transparency and reduced fraud through automated fare systems in Kigali's public transport network. These findings align with the goal of your study, which aims to improve Ceres Liner's ticketing process by automating ticket bookings and integrating electronic payment options like cash and GCash.

Locally, studies by Riah Encarnacion (2024) and Galang et al. (2025) further reinforce the positive impact of digital ticketing systems on transportation operations in the Philippines. Mejas & Encarnacion's development of a web-based ticketing system for maritime travel in Siargao Island illustrates the advantages of online booking in reducing operational inefficiencies, especially in terms of managing ticket sales and improving passenger experience. Similarly, Galang et al. examined the influence of ride-hailing services and digital platforms on urban mobility in Pampanga, highlighting that digital systems not only reduce commuting time but also enhance fare collection and payment transparency. Both studies show that digital systems contribute to faster service and smoother transaction processes, which are essential for improving Ceres Liner's operations.

In conclusion, the synthesis of these studies shows that the adoption of digital ticketing systems leads to improved efficiency, greater customer satisfaction, and better data management. These findings support the development of a similar system for Ceres Liner in Negros Occidental, which aims to streamline its ticketing process, reduce errors, and introduce flexible payment options. The research highlights the growing role of technology in transforming public transportation services, demonstrating how digital solutions can drive improvements in both service delivery and operational management.

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